

Tutorial Sheet-04

1. Let X be a random variable such that

$$n^a \mathbb{P}(|X| > n) \rightarrow 0 \quad \text{as } n \rightarrow \infty$$

for some $a > 0$. Then Prove that $\mathbb{E}(|X|^p) < \infty$ for all $0 < p < a$.

2. Let X be a random variable with $\mathbb{E}(X) = 0$ and $\text{Var}(X) = \sigma^2$. Then Prove that for $x > 0$,

$$\mathbb{P}(X \geq x) \leq \frac{\sigma^2}{\sigma^2 + x^2},$$

and for $x < 0$,

$$\mathbb{P}(X \geq x) \geq \frac{x^2}{\sigma^2 + x^2}.$$

3. Let X be a random variable such that $\mathbb{E}(|X|^4) < \infty$ and suppose that

$$\mathbb{E}(X) = 0, \quad \mathbb{E}(X^2) = \sigma^2.$$

Then Prove that for all $K > 1$,

$$\mathbb{P}(|X| > K\sigma) \leq \frac{\mu_4 - \sigma^4}{\mu_4 + K^4\sigma^4 - 2K^2\sigma^4},$$

where $\mu_4 = \mathbb{E}(X^4)$.

4. Let X be a random variable with PDF

$$f_X(x) = \begin{cases} \frac{x^A e^{-x}}{\Gamma(A+1)}, & x > 0, \\ 0, & \text{otherwise,} \end{cases}$$

where $A > 0$ is an integer. Show that

$$\mathbb{P}(0 < X < 2(A+1)) > \frac{A}{A+1}.$$

5. Use Chebyshev's inequality to show that for any $k > 1$,

$$e^{k+1} \geq k^2,$$

.

6. For any random variable X , show that

$$\mathbb{P}(X > 0) \leq \inf_{t>0} \varphi(t) \leq 1,$$

where

$$\varphi(t) = \mathbb{E}(e^{tX}), \quad t > 0,$$

and $0 < \varphi(t) \leq \infty$.

7. Let (X, Y) be random variable with joint probability density function

$$f_{X,Y}(x, y) = \frac{c}{2\pi (c^2 + x^2 + y^2)^{3/2}}, \quad -\infty < x < \infty, -\infty < y < \infty,$$

where $c > 0$. Find the marginal probability density functions of X and Y .

8. Let (X, Y) have the distribution with joint PMF

$$\mathbb{P}(X = x, Y = y) = \frac{(x + y + k - 1)!}{x! y! (k - 1)!} p_1^x p_2^y (1 - p_1 - p_2)^k,$$

where

$$x, y = 0, 1, 2, \dots, \quad k > 1 \text{ is an integer,} \quad 0 < p_1 < 1, \quad 0 < p_2 < 1, \quad p_1 + p_2 < 1.$$

Find the marginal PMFs of X and Y .

9. Is the function

$$f_{X,Y,Z,U}(x, y, z, u) = \begin{cases} e^{-u}, & 0 < x < y < z < u < \infty, \\ 0, & \text{elsewhere,} \end{cases}$$

a joint probability density function? If so, find $\mathbb{P}(X < 7)$, where (X, Y, Z, U) is a random vector with joint density $f_{X,Y,Z,U}$.

10. Show that the function

$$f_{X,Y,Z,U}(x, y, z, u) = \begin{cases} \frac{24}{(1 + x + y + z + u)^5}, & x > 0, y > 0, z > 0, u > 0, \\ 0, & \text{elsewhere,} \end{cases}$$

is a joint probability density function.

- (a) Find $\mathbb{P}(X > Y > Z > U)$.
- (b) Find $\mathbb{P}(X + Y + Z + U > 1)$.

11. Let (X, Y) be a continuous jointly random variable with joint PDF $f(x, y)$ that is continuous everywhere. What is the value of $P(X = Y)$? Explain geometrically why this result holds for continuous variables but not necessarily for discrete variables.
12. Let (X, Y) be uniformly distributed over the unit disk $D = \{(x, y) : x^2 + y^2 \leq 1\}$. The joint PDF is $f(x, y) = C$ for $(x, y) \in D$ and 0 otherwise.
- (a) Find the constant C .
 - (b) Find the Marginal PDF $f_X(x)$. (Note the limits of integration depend on x).
 - (c) Are X and Y independent? Justify.
13. Let X and Y have joint PDF $f(x, y) = 4xy$ for $0 < x < 1, 0 < y < 1$. Find the probability $P(Y/X > 1)$.